Q 1:Write a C program that acts as a simple calculator. The program should take two numbers and an operator as input from the user and perform the respective operation (addition, subtraction, multiplication, division, or modulus) using operators. Challenge: Extend the program to handle invalid operator inputs.

A.
```
#include<stdio.h>
 main(){
    int a,b;
    printf("\n enter the value of a & b");
    scanf("%d%d",&a,&b);
    printf("\n add=%d",a+b);
    printf("\n sub=%d",a-b);
    printf("\n mul=%d",a*b);
    printf("\n div=%f",(float)a/b);
    printf("\n mod=%d",a%b);
    return 0;

}
```

Q 2:Write a C program that takes an integer from the user and checks the following using different operators:
- **o Whether the number is even or odd.**
- **o Whether the number is positive, negative, or zero.**
- **o Whether the number is a multiple of both 3 and 5.**

A. o
```
#include<stdio.h>
main(){
    int num;
    printf("\n enter num : ");
    scanf("%d" ,&num);
    if(num%2==0){
        printf("\n number is even");
    }
    else{
        printf("\n number is odd");
    }
   return 0;
}
```

o. #include<stdio.h>
```
main(){
   int num;
   printf("\n enter num");
   scanf("%d" ,&num);
   if(num>0){
      printf("\n The number is +ve");
   }
   else if(num<0){
      printf("\n The number is -ve");
   }
   else{
      printf("\n The number is zero.");
   }
 return 0;
}
```

o. #include <stdio.h>
```
   int main() {
   int num;
   printf("Enter an integer: ");
   scanf("%d", &num);
      if (num % 3 == 0 && num % 5 == 0) {
      printf("%d is a multiple of both 3 and 5.\n", num);
   } else {
      printf("%d is NOT a multiple of both 3 and 5.\n", num);
   }
   return 0;
}
```

Q 3: Write a C program that takes the marks of a student as input and displays the corresponding grade based on the following conditions:

- **o Marks > 90: Grade A**
- **o Marks > 75 and <= 90: Grade B**
- **o Marks > 50 and <= 75: Grade C**
- **o Marks <= 50: Grade D**

Use if-else or switch statements for the decision-making process.

A. #include <stdio.h>
```
   int main(){
     int marks;
```

```
printf("\n Enter the marks of the student");
scanf("%d" ,&marks);
if(marks>90){
    printf("\n Grade A");
}
else if(marks>75 && marks<=90){
    printf("\n Grade B");
}
else if(marks>50 && marks<=75){
    printf("\n Grade C");
}
else{
   printf("\n Grade D");
}
 return 0;
}
```

Q 4:Write a C program that takes three numbers from the user and determines:

- **o The largest number.**
- **o The smallest number.**

Challenge: Solve the problem using both if-else and switch-case statements.

A. 1.<u>Using if-else statement</u>

```
#include <stdio.h>

int main() {
int a, b, c;
printf("Enter three numbers: ");
scanf("%d %d %d", &a, &b, &c);

int largest;
if (a >= b && a >= c)
   largest = a;
else if (b >= a && b >= c)
   largest = b;
else
   largest = c;
```

```
    int smallest;
     if (a <= b && a <= c)
        smallest = a;
    else if (b <= a && b <= c)
        smallest = b;
    else
        smallest = c;

    printf("Largest number = %d\n", largest);
    printf("Smallest number = %d\n", smallest);

    return 0;
}
```

2.Using switch-case statement

```
#include <stdio.h>

int main() {
    int a, b, c;
    int largest, smallest;
    printf("Enter three numbers: ");
    scanf("%d %d %d", &a, &b, &c);
    largest = (a > b ? (a > c ? a : c) : (b > c ? b : c));
    smallest = (a < b ? (a < c ? a : c) : (b < c ? b : c));
    printf("Largest number = ");
    switch(1) {
        case 1:
            printf("%d\n", largest);
            break;
    }
    printf("Smallest number = ");
    switch(2) {
        case 2:
            printf("%d\n", smallest);
            break;
     }

       return 0;
}
```

Q 5:Write a C program that checks whether a given number is a prime number or not using a for loop.
Challenge: Modify the program to print all prime numbers between 1 and a given number.

A.
```
#include <stdio.h>
 int main() {
 int n, i;
 printf("Enter a number: ");
 scanf("%d", &n);
 if (n <= 1) {
    printf("%d is not a prime number.\n", n);
 }
 else {
 for (i = 2; i <= n - 1; i++) {
        if (n % i == 0) {
 printf("%d is a prime number.\n", n);
}
return 0;
}
```

MODIFICATION

```
#include <stdio.h>
   int main() {
   int n, i, j, flag;
   printf("Enter a number: ");
   scanf("%d", &n);
   printf("Prime numbers between 1 and %d are:\n", n);
   for (i = 2; i <= n; i++) {
      flag = 0;
      for (j = 2; j <= i - 1; j++) {
         if (i % j == 0) {
            flag = 1;
            break;
         }
      }
      if (flag == 0)
         printf("%d ", i);
   }
   return 0;
}
```

Q 6:Write a C program that takes an integer input from the user and prints its multiplication table using a for loop.

A.
```
#include<stdio.h>
   main(){
   int mul,num;
   printf("Enter a number : ");
   scanf("%d",&num);
   for(mul=1;mul<=10;mul++){
   printf("\n%d * %d = %d",num,mul,num*mul);
}
}
```

Q 7:Write a C program that takes an integer from the user and calculates the sum of its digits using a while loop.

Challenge: Extend the program to reverse the digits of the number.

A.
```
#include<stdio.h>
   main(){
   int sum=0,rem,num,r;
   printf(" Enter a number:");
   scanf("%d",&num);
   r=num;
   while(r!=0){
      rem=r%10;
      r=r/10;
      sum=sum+rem;
   }
   printf(" Sum of digits: %d",sum);
}
```

<u>MODIFICATION</u>

```
#include<stdio.h>
main(){
  int num,rem,rev=0;
  printf("\n Enter num: ");
  scanf("%d",&num);
  while(num!=0){
    rem=num%10;
    num=num/10;
    rev=rev*10+rem;
```

```
   }
  printf("\t rev=%d",rev);
}
```

Q 8:Write a C program that accepts 10 integers from the user and stores them in an array. The program should then find and print the maximum and minimum values in the array.

Challenge: Extend the program to sort the array in ascending order.

A.
```
#include<stdio.h>
     main(){
     int i,min,max=0,arr[10];
     for(i=0;i<10;i++){
     printf("\n Enter array elements :a[%d]",i);
     scanf("%d",&a[i]);
        if(min>arr[i]){
            min=arr[i];
        }
     }
     for(i=0;i<10;i++){
        if(max<arr[i]){
            max=arr[i];
        }
    }
    printf("\n minimum element of array is=%d",min);
    printf("\n maximum element of array is=%d",max);

 }
```

<u>Modification</u>

```
#include<stdio.h>
int main(){
    int a[5];
    int i,j,temp;
    for(i=0;i<5;i++){
        printf("\n Enter value for a[%d]:",i);
        scanf("%d",&a[i]);
    }
    //sort
    for(i=0;i<5;i++){
        for(j=i+1;j<5;j++){
```

```
     if(a[i]<a[j]){
              temp=a[i];
              a[i]=a[j];
              a[j]=temp;
           }
        }
    }
    printf("\n Sorted array elements are:");
    for(i=0;i<5;i++){
       printf("\n a[%d]=%d",i,a[i]);
    }
}
```

Q 9: Write a C program that accepts two 2x2 matrices from the user and adds them. Display the resultant matrix.
Challenge: Extend the program to work with 3x3 matrices and matrix multiplication.

A.

```
#include<stdio.h>
int main(){
    int a[2][2],b[2][2],c[2][2];
    int i,j;
    printf("\n Enter array elements of a :");
    for(i=0;i<2;i++){
       for(j=0;j<2;j++){
          printf("\n a[%d][%d]:",i,j);
          scanf("%d",&a[i][j]);
       }
    }
    printf("\n array elements \n");
    for(i=0;i<2;i++){
       printf("|");
       for(j=0;j<2;j++){
          printf("\t a[%d][%d]=%d",i,j,a[i][j]);
       }
       printf("|\n");
    }
}
printf("\n Enter array elements of b :");
    for(i=0;i<2;i++){
       for(j=0;j<2;j++){
          printf("\n b[%d][%d]:",i,j);
```

```
            scanf("%d",&a[i][j]);
        }
    }
    printf("\n array elements \n");
    for(i=0;i<2;i++){
        printf("|");
        for(j=0;j<2;j++){
            printf("\t b[%d][%d]=%d",i,j,b[i][j]);
        }
        printf("|\n");
    }
}
for (i = 0; i < 2; i++)
    {
        for (j = 0; j < 2; j++)
        {
            c[i][j] = a[i][j] + b[i][j];
        }
    }
    printf("\n Addition of Matrix-A and Matrix-B");
    printf("\n Array elements of c \n");
    for (i = 0; i < 2; i++)
    {
        printf("|");
        for (j = 0; j < 2; j++)
        {
            printf("\t c[%d][%d]=%d", i, j, c[i][j]);
        }
        printf("|\n");
    }
}
```

Q 10:Write a C program that takes N numbers from the user and stores them in an array. The program should then calculate and display the sum of all array elements.

A.
```
#include<stdio.h>
    main(){
    int n,a[n],sum=0;
    printf(“\n Enter number of array elements :”);
        scanf(“%d”,&n);
    int i;
```

```
 for(i=0;i<n;i++){
    printf("\n Enter value for a[%d]:",i);
    scanf("%d",&a[i]);
    sum=sum+a[i];
 }
 //Display values of array
 for(i=0;i<n;i++){
    printf("\n a[%d]=%d",i,a[i]);
 }
 printf("\n Sum of array elements=%d",sum);
}
```

Q 11:Write a C program that generates the Fibonacci sequence up to N terms using a recursive function.

A.

```
#include <stdio.h>

int fib(int n) {
   if (n == 0)
      return 0;
   else if (n == 1)
      return 1;
   else
      return fib(n - 1) + fib(n - 2);
}
int main() {
   int n, i;

   printf("Enter number of terms: ");
   scanf("%d", &n);

   printf("Fibonacci sequence:\n");

   for (i = 0; i < n; i++) {
      printf("%d ", fib(i));
   }
   return 0;
}
```

Q 12:Write a C program that calculates the factorial of a given number using a function.

A.

```
#include<stdio.h>
```

```
int factFind(int num){
   int f;
   if(num==1){
      return 1;
   }
   f=num*factFind(num-1);
   return f;
}
main(){
   printf("\n factorial=%d",factFind(num));
}
```

Q 13:Write a C program that takes a number as input and checks whether it is a palindrome using a function.

Challenge: Modify the program to check if a given string is a palindrome.

A.
```
#include <stdio.h>

   int Palindrome(int n) {
   int rev = 0, r = n;

   while r != 0) {
      rev = rev * 10 + r % 10;
      r = r / 10;
   }
   if (rev == n)
      return 1;
   else
      return 0;
}
int main() {
   int num;
   printf("Enter a number: ");
   scanf("%d", &num);
   if (Palindrome(num))
      printf("%d is a palindrome number.\n", num);
   else
      printf("%d is not a palindrome number.\n", num);

   return 0;
}
```

MODIFICATION

```
#include <stdio.h>
#include <string.h>
int StringPalindrome(char str[]) {
    int i = 0;
    int j = strlen(str) - 1;
    while (i < j) {
        if (str[i] != str[j])
            return 0;
        i++;
        j--;
    }
    return 1;
}
int main() {
    char str[100];
    printf("Enter a string: ");
    scanf("%s", str);
    if (StringPalindrome(str))
        printf("The string is a palindrome.\n");
    else
        printf("The string is not a palindrome.\n");
    return 0;
}
```

Q 14:Write a C program that takes a string as input and reverses it using a function.

A.

```
#include<stdio.h>
    #include<string.h>
    int main(){
    char str[20]
    printf("\n Enter The str:");
    scanf("%s",str);
    strrev(str);
   printf("\n reverse of string is %s ",strrev(str));
   return 0;
}
```

Q 15:Write a C program that takes a string from the user and counts the number of vowels and consonants in the string.

Challenge: Extend the program to also count digits and special characters.

A.
```
#include <stdio.h>
   #include <ctype.h>
   int main() {
   char str[200];
   int vowels = 0, consonants = 0, digits = 0, special = 0;
   int i;
   printf("Enter a string: ");
   gets(str);
   puts(str);
    while(str[i]!='\0'){
        if (isalpha(ch)) {
          ch = tolower(ch);
              if (ch == 'a' || ch == 'e' || ch == 'i' ||
             ch == 'o' || ch == 'u')
             vowels++;
          else
             consonants++;
       }
     else if (isdigit(ch)) {
          digits++;
       }
       else if (ch != '\n') {
          special++;
       }
   }
   printf("\nVowels: %d", vowels);
   printf("\nConsonants: %d", consonants);
   printf("\nDigits: %d", digits);
   printf("\nSpecial characters: %d\n", special);
   return 0;
}
```